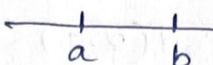
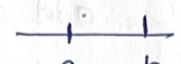


Data Analytics :- OPEN & CLOSED

Open sphere (Ball) / closed sphere (Ball) in metric space.

Closed :-  $[a, b]$
 $\downarrow \quad \downarrow$
 boundary points
 a, b included

Open Interval :-  (a, b)
 (a, b) are not included.

Open sphere

Let (X, d) be metric space.



Let $a \in X$ and $r > 0$ be any real no.
element of X. radius.

then set $S_r(a) = \{x \in X; d(a, x) < r\}$ is called open sphere in space.
 $\downarrow$
distance

closed sphere

Let (X, d) be metric space.

Let $a \in X$ and $r > 0$ be any real no.

then set $S_r[a] = \{x \in X; d(a, x) \leq r\}$ is called closed sphere.
 $S_r[a], S[a, r], B_r[a], B[a, r]$

$$-y_2^2 \leq y_2$$

$$\frac{x_2 y_2}{2} < x_2^2$$

$$(x_2)^2 < x_2^2$$

$$1 - y_2^2 < x_2^2$$

$$1 - y_2^2 \leq x_2^2$$

ge³

ka A neigh
b hoga

Open and closed sphere for usual metric space in $\mathbb{R}$.

$d(x, y) = |x - y|$ $x, y \in \mathbb{R}$. $(\mathbb{R}, d)$ make metric space.

(1) open sphere $S_r(a) = \{x \in X : d(x, a) < r\}$

sphere

$$\Rightarrow \{x \in \mathbb{R} : d(x, a) < r\} \quad \left\{ \begin{array}{l} |x| < c \\ -c < x < c \end{array} \right.$$

$$\Rightarrow \{x \in \mathbb{R} : |x - a| < r\}$$

$$\Rightarrow \{x \in \mathbb{R} : -r < x - a < r\}$$

add a to all

$$\{x \in \mathbb{R} : \underline{a - r} < x < \underline{a + r}\}$$

$\mathbb{R}$ element r

$$\Rightarrow \{x \in \mathbb{R} : (\underline{a - r}, \underline{a + r})\}$$

open Interval

(2) closed sphere

$$S_r[a] = \{x \in X : d(x, a) \leq r\}$$

$$\Rightarrow \{x \in \mathbb{R} : |x - a| \leq r\}$$

$$\Rightarrow \{x \in \mathbb{R} : -r \leq x - a \leq r\}$$

$$\Rightarrow \{x \in \mathbb{R} : \underline{a - r} \leq x \leq \underline{a + r}\}$$

$$S_r[a] = \underline{[a - r, a + r]}$$

closed Interval

metric space?
 (R, d) make
 metric

$$\begin{cases} |x| < c \\ -c < x < c \end{cases}$$

let
 be

Rel.

eg

open & closed sphere in R^2 'd'

d - Euclidean distance.

$$(z_1)(z_2) \in R^2 \quad d(z_1, z_2) = [(x_1 - x_2)^2 + (y_1 - y_2)^2]^{1/2}$$

$$z_1 = (x_1, y_1)$$

$$z_2 = (x_2, y_2)$$

$$(i) \text{ open sphere :- } S_r(a) = \{ x \in R^2 : d(\overset{x_1, y_1}{x}, \overset{x_2, y_2}{a}) < r \}$$

$$\Rightarrow \{ x \in R^2 : \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} < r \}$$

$$\Rightarrow \{ x \in R^2 : (x_1 - x_2)^2 + (y_1 - y_2)^2 < r^2 \}$$

(2) closed sphere:-

$$S_r[a] = \{ x \in R^2 : (x_1 - x_2)^2 + (y_1 - y_2)^2 \leq r^2 \}$$

Neighbourhood of a point in metric space

(basically a set)

kise set ko pt. ka neighbourhood kab kahenge?

$$A \subseteq (X, d)$$

$$\boxed{a, r > 0}$$

center like make
 an open sphere.

$$a \in S_r(a) \subseteq A$$

$\therefore A$ is nbd of pt a .

$$\boxed{a \in A}$$

also if $a \rightarrow A$ nbd then

$a \rightarrow$ interior pt of A .

'a' ka A neigho
 kab hoga.

Formal Definition

$$A \subseteq (X, d)$$

Let $A \subseteq (X, d)$, then a pt $a' \in A$ is called Interior point of A if (a') is the center of an open sphere contained in A .

i.e. $\exists r > 0$ such that $S_r(a') \subseteq A$.

A is said to be neighbourhood of point (a') .

eg $(0, 2)$ $1 \in (0, 2)$

check 1 is int pt of $(0, 2)$.

$r = 1/2$ open sphere

$$a = 1, r = 1/2$$

open sphere $(a-r, a+r)$

$$(1-1/2, 1+1/2)$$

$$(1/2, 1.5) \subseteq (0, 2)$$

Now check $0 \in (0, 2)$?

check 0 is nbd of $(0, 2)$.

0 does not belongs to $(0, 2)$

$$0 \notin (0, 2)$$

eg $0 \in [0, 2]$ check.

$$0 \in [0, 2]$$

$$\epsilon > 0 \quad (a-r, a+r)$$

$$(0-\epsilon, 0+\epsilon)$$

$$(-\epsilon, \epsilon)$$

does not nbd of $[0, 2]$

of A

all its pts.

Then A is said to

such that

$$s = (1, 2)$$

(because it is nbd of all pt $(1, 2)$)

is not Interior A

are open set

pts $\checkmark$
to in ϕ_1
stied)

1. metric space?

A is called Interior of an open sphere

point (a).

OPEN SET

$$A \subseteq (X, d)$$

If every pt of A is interior pt of A

i.e; A is neighbourhood of all its pts.

Let (X, d) be metric space & $A \subseteq X$, Then A is said to be open if for every $a \in A$, $\exists \delta > 0$ such that

$$S_\delta(a) \subseteq A$$

eg (R, d) be metric space

$$(1, 2) \subseteq (R, d)$$

↑ open set.

$$[(1, 2)]^o$$

interior set of (1,2) is

$$= (1, 2)$$

$\Rightarrow (1, 2) \leftarrow$ open set (because it is nbd of all pt (1,2))

$$[1, 2] \subseteq (R, d)$$

↑ not an open set.

$$[1, 2]^o = (1, 2)$$

↑ except.

$$(1-\epsilon, 1+\epsilon) \text{ cover } 1 \text{ is not interior}$$

$$(2-\epsilon, 2+\epsilon)$$

2 is not interior A.

In any metric space X , ϕ & X are open set

$$\phi \subseteq (X, d)$$

$$\phi \subseteq X$$

empty

↳ open.

ϕ is open $\because$ it is nbd of all its pts ✓

(Since there are not points in ϕ)

So requirement is satisfied)

for usual metric space in

$$X \subseteq X, x \in X \quad \exists r > 0 \text{ such that } S_r(x) \subseteq X \checkmark$$

pt $a' \in A$ is closed.

the centre of an

that.

$$S_r(a) \subseteq A.$$

hood of point

Q. $S_r(x) \subseteq A$
 $p \in S_r(x)$
Is A nbd of pt p ?

$$p \in S_r(x) \\ d(x, p) < r.$$

$$\boxed{r_1 = r - d(x, p) > 0}$$
$$r_1 > 0.$$

$$S_{r_1}(p) \subseteq A$$

$$S_{r_1}(p) \subseteq S_r(x) \checkmark \text{ Trick.}$$

$$y \in S_{r_1}(p) \Rightarrow y \in S_r(x)$$

arbitrary
element.

$$y \in S_{r_1}(p)$$

$$d(y, p) < r_1$$

$$d(y, p) < r - d(x, p)$$

$$d(x, p) + d(y, p) < r$$

$$d(x, p) + d(p, y) < r$$

$$d(x, p) + d(p, y) < d(x, y)$$

$$d(x, y) < r$$

$$\Rightarrow y \in S_r(x)$$

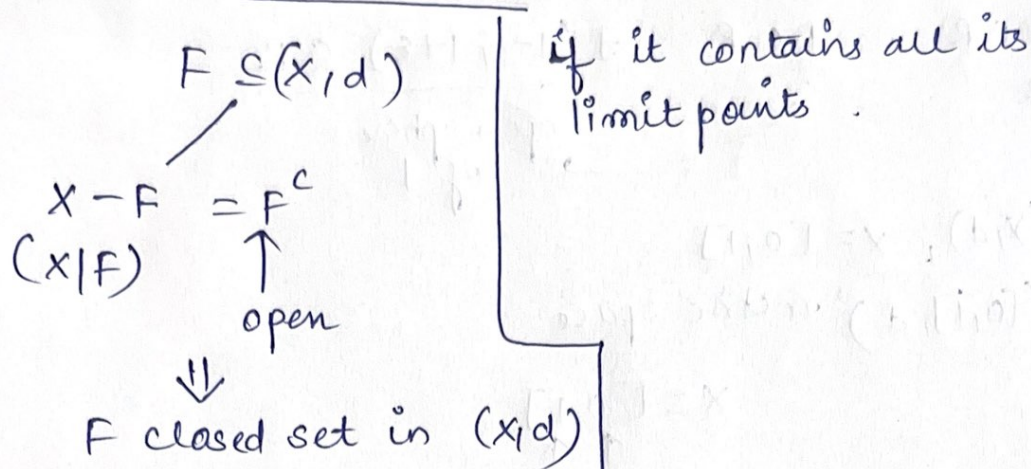
$X \subseteq X$, $x \in X \quad \exists \delta > 0$ such that $S_\delta(x) \subseteq X$ ✓

$\xrightarrow{x=R} R \subseteq R$, $1 \in R \quad \exists (1-\epsilon, 1+\epsilon) \subseteq R$
open sphere of 1

(X, d) , $X = [0, 1]$
 $([0, 1], d)$ metric space.

$X = [0, 1]$
 $\uparrow \quad \uparrow$
 $(0-\epsilon, 0+\epsilon)$
 $[0, \epsilon) \subseteq [0, 1]$
 $\uparrow$ open sphere.

Closed set



① $\frac{[a, b]}{F} \subseteq (R, d)$

$\frac{X - F}{R - [a, b]} \xleftrightarrow[\leftarrow]{\rightarrow} \frac{[a, b]}{a \quad b} \xleftrightarrow[\leftarrow]{\rightarrow} \infty$

$(-\infty, a) \cup (b, \infty)$
 open open interval
 (interval) i.e. open set] open.

$[a, b]^c \rightarrow \text{open}$

$\therefore$ it is closed set in (R, d)